

Jeff Decary

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EDUCATION

University of Connecticut

Storrs, CT, United States

Ph.D. Candidate in Business Administration

Aug. 2020 – Present

- Concentration: *Operations and Information Management* (OPIM)

Polytechnique Montreal

Montreal, QC, Canada

M.Sc.A. Mathematics

Aug. 2018 – Sept. 2020

- Member of the Canada Excellence Research Chair in Data Science for Real-Time Decision-Making

University of Quebec in Montreal

Montreal, QC, Canada

B.Sc.(Honors) Actuarial Science

Aug. 2015 – May 2018

WORKING PAPERS

Major Revision | Production and Operations Management

February 2025

- Decary, Jeff., Bergman, David, Cardonha, Carlos, Imbrogno, & Jason, Lodi, Andrea. **“The Madness of Multiple Entries in *March Madness*”**.
 - Developed heuristics using mixed-integer linear programming to optimize March Madness entries
 - Our solution outperforms all 12,605 entrants in a DraftKings contest in terms of expected score
 - Achieves a 2.02% chance of winning \$1M on a \$10K investment—the highest among all entrants

Reject and Resubmit | Operations Research

February 2025

- Decary, Jeff, Bergman, David, & Zou, Bin.
“Practical Portfolio Optimization for Parlay Betting on Simultaneous Events”
 - Introduced a logic-based Benders decomposition method for practical betting optimization
 - Achieved a 445% expected wealth increase over six days of real-world betting
 - Economic analysis of parlay pricing from the sportsbook perspective.

Work in Progress

- Decary, Jeff, Bergman, David, & Day, Robert. **“Pricing Mechanism for Daily Fantasy Sports.”**
- Shi, Chenbo, Decary, Jeff, Qu, Mengwei, Emadikhiav, Mohsen, Bergman, David. **“Robust Treatment Effect Estimation via Randomized Matching on Convex Bipartite Graph”**
- Decary, Jeff, Wang, Keliang, Cardonha, Carlos, Bergman, David, Bai, Miao. **“Branch-and-Price Algorithms for Embedded Neural Network Optimization: Application on a Surgical Room Scheduling Problem”**

TEACHING EXPERIENCE

Instructor of Record

Aug. 2020 – Present

University of Connecticut

Storrs, USA

- **OPIM 3103 (Fall 2022-Spring 2023):**
 - Designed the syllabus and delivered lectures, introducing Excel and business information systems through hands-on data analysis projects.
 - Managed grading and student support for 40 undergraduates
 - Received an average score of **4.0/5.0** for “Instructor Overall Effectiveness”

Teaching Assistant

University of Connecticut

Aug. 2020 – Present

Storrs, USA

- **OPIM 3602 (Spring 2024)**: Graded assignments and exams; held office hours for advanced analytics course
- **OPIM 3104 (Spring 2024)**: Delivered a guest lecture on machine learning applications
- **OPIM 3104 (Fall 2021, Spring 2024)**: Led in-class activities, graded exams, and held office hours
- **OPIM 3103 (Fall 2020, Spring 2021, Spring 2022)**: Assisted with grading assignments and exams

INDUSTRY EXPERIENCE

Asset Management and Pensions Intern

Normandin Beaudry

May 2017 – Aug. 2017

Montreal, Canada

- Performed actuarial valuations and long-term financial projections for pension funds
- Prepared client-facing asset review presentations and supported investment strategy discussions
- Gained practical insight into institutional investment strategy and retirement plan design

Business Intelligence Intern

The Co-operators

Jan. 2016 – Jul. 2016

Montreal, Canada

- Reconciled legacy systems and analyzed insurance databases to support system migration
- Contributed to internal reporting solutions through database analysis and business logic design

AWARDS AND FELLOWSHIPS

Summer Doctoral Dissertation Fellowship, *University of Connecticut*

May 2025

OPIM Department Summer Fellowship, *University of Connecticut School of Business*

May 2024

Named Scholar-Scholarship, *University of Connecticut School of Business*

May 2023

UConn School of Business Dean's Summer Fellowship, *University of Connecticut*

May 2022-2024

Predoctoral Summer Fellowship, *University of Connecticut School of Business*

May 2021-2024

COMPETITIONS

Sixth place in *March Machine Learning Mania 2024* organized by *Kaggle*

April 2024

- Developed predictive models for NCAA outcomes; placed 6th in international Kaggle competition

Third place at Munich Re Cup

Jan. 2018–April 2018

- The Munich Re Cup is a case competition open to the Centers of Actuarial Excellence in Canada.
- Developed an accelerated underwriting program using machine learning
- Presented to senior executives of Munich Re

PROFESSIONAL CERTIFICATION

Actuarial Certification – Society of Actuaries

2015-2017

- Passed: Probability, Financial Mathematics, Models for Financial Economics, Construction of Actuarial Models

VOLUNTEERING AND LEADERSHIP

Organizer Student-Led Seminar Series

2024-2025

- Led the planning and coordination of seminars on Business Analytics research and academic career development
- Facilitated discussions on key milestones in the path to becoming a professor

Conference Volunteer, CP2023

Aug.-Sept. 2023

- Co-organized the Workshop: **Optimization for Sports**
- General Assistance

Technical Support, CP2021

Oct. 2021

Secretary, Board of Directors, *Tennis de Table Québec*

May 2019-June 2023

- Oversaw financial and strategic planning for a \$500,000 annual budget
- Secured \$46,000 in funding from the Government of Québec to develop a video analysis tool through computer vision and data science

LANGUAGES, SKILLS, AND INTEREST

- **Languages:** French & English
- **Technical Skills:** Python, R, Julia, MatLab, Pytorch, Gurobi and L^AT_EX
- **Interests:** Racket sports, team sports, board games, cooking, travel